Milestone1

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Using simulated data modelled on OECD unemployment statistics, this project plans to compare tertiary-educated foreign-born and native-born women across more and less developed regions, and by age at arrival.

Unemployment among Highly Educated Immigrant Women by Destination and Age at Arrival

Introduction

Questions

Immigrants with a university degree are often expected to secure stable positions in the labor market, yet a degree does not guarantee the same outcome across nations. Immigrants may face disadvantages that combine gender and migration status. The strength of these disadvantages may depend on where they settle and on when they arrived. For this research, I focus on highly educated female immigrants and ask: **Does the unemployment rate of tertiary-educated foreign-born women vary by the development level of the destination country and by age at arrival?**

First, **the destination country**: as more developed countries may have more established systems for recognizing foreign qualifications, I expect highly educated foreign-born women to experience lower unemployment in Northern America, Oceania, Western Europe and Northern Europe, which I group as "more developed" regions, than in Latin America, Southern Europe and Eastern Europe, which I group as "less developed" regions. Second, **age at arrival**: women who arrived before age 16 were likely to have integrated and completed much of their schooling in the destination country, so I expect their unemployment to be lower than that of women who arrived at age 16 or older, whose qualifications may have been earned elsewhere and who may have had less time to become familiar with the destination country.

To make the analysis more robust, I also account for general labor market conditions. A low unemployment rate among immigrants may simply reflect a strong labor market that benefits everyone. To separate the two, I also compare each foreign-born group with native-born women of the same region and education level, and examine the gap between them (foreign-born minus

native-born unemployment rate). If immigrant women are better off in more developed regions for reasons beyond general labor market conditions, I expect this gap to be smaller in more developed regions than in less developed ones.

Data

The analysis was conducted in R (R Core Team 2025) (Alexander 2023). For this project, I plan to using **2023 OECD data on unemployment rates** (OECD 2023). The data come from the OECD Data Explorer (dataflow DSD_EAG_LSO_EA@DF_LSO_NEAC_UNEMP_MIGR). Each row is a subgroup in one country, defined by sex, age group, education level, place of birth (foreign-born or native-born) and, for the foreign-born, age at arrival (before age 16, or 16 and older). The outcome variable is the unemployment rate, measured as **a percentage of the subgroup's labor force**. I plan to keep women with tertiary education, group countries into regions, and then into "more developed" and "less developed" regions. I compare foreign-born women **by region** and **by age at arrival**, and compare them with native-born women of the same region and education level. I exclude Israel because it is difficult to classify as either a more or less developed region, and I exclude the OECD average because it is not a country.

Limitations

Although the OECD provides detailed data, it does not report immigrants' country of origin. Where immigrants come from may affect whether their educational credentials are recognized in the country of destination, and therefore their unemployment rate. As a result, differences across destination countries in this analysis may partly reflect differences in the origins of their immigrant populations, rather than characteristics of the destination countries themselves.

Expected Results

I will treat the hypotheses as supported if three conditions hold. First, for both arrival-age groups, the mean unemployment rate is lower in more developed than in less developed regions by about 1 percentage point or more. Second, within each development group, women who arrived before age 16 have a lower mean unemployment rate than women who arrived at age 16 or older by about 1 percentage point or more. Third, for both arrival-age groups, the gap between foreign-born and native-born women (foreign-born minus native-born) is smaller in more developed regions than in less developed regions by about 1 percentage point or more. If the differences are small, or are inconsistent across groups, I will conclude that there is no clear pattern.

```r
library(tidyverse)
library(tinytex)
library(knitr)
```

```r
raw_set <- read_csv(
  "OECD_raw_v2.csv",
  show_col_types = FALSE
)
```

PART I: Simulating Data

```r
set.seed(916)

#Simulate the structure of the real data, based on the data of OECD
country_region <- tibble(
  country = c("Canada", "United States",
              "Mexico", "Costa Rica", "Colombia", "Chile",
              "Australia", "New Zealand",
              "Austria", "Belgium", "France", "Germany", "Luxembourg",
              "Netherlands", "Switzerland",
              "Estonia", "Ireland", "Latvia", "Lithuania", "Norway",
              "Sweden", "United Kingdom",
              "Greece", "Italy", "Portugal", "Slovenia", "Spain",
              "Bulgaria", "Czechia", "Hungary", "Poland", "Romania",
              "Slovak Republic"),
  region  = c(rep("Northern America", 2),
              rep("Latin America", 4),
              rep("Oceania", 2),
              rep("Western Europe", 7),
              rep("Northern Europe", 7),
              rep("Southern Europe", 5),
              rep("Eastern Europe", 6))
)

head(country_region)
```

```
# A tibble: 6 x 2
```

```
  country       region
  <chr>         <chr>
1 Canada        Northern America
2 United States Northern America
3 Mexico        Latin America
4 Costa Rica    Latin America
5 Colombia      Latin America
6 Chile         Latin America
```

```
#add characteristics
n_countries <- length(country_region$country)

simulated_ppl <- tibble(
  country = rep(country_region$country, each = 18),
  sex = rep(rep(c("Female", "Male"), each = 9), times = n_countries),
  education = rep(rep(c("Below upper secondary",
                        "Upper secondary / post-secondary non-tertiary",
                        "Tertiary"), each = 3),
                  times = 2 * n_countries),
  birth_place = rep(c("Foreign-born", "Foreign-born", "Native-born"),
                    times = 6 * n_countries),
  arrival_age = rep(c("Arrived before age 16", "Arrived at age 16 or older",
                      "Not applicable"),
                    times = 6 * n_countries)
)

head(simulated_ppl)
```

```
# A tibble: 6 x 5
  country sex    education                                birth_place arrival_age
  <chr>   <chr>  <chr>                                    <chr>       <chr>
1 Canada  Female Below upper secondary                    Foreign-bo~ Arrived be~
2 Canada  Female Below upper secondary                    Foreign-bo~ Arrived at~
3 Canada  Female Below upper secondary                    Native-born Not applic~
4 Canada  Female Upper secondary / post-secondary non-t~ Foreign-bo~ Arrived be~
5 Canada  Female Upper secondary / post-secondary non-t~ Foreign-bo~ Arrived at~
6 Canada  Female Upper secondary / post-secondary non-t~ Native-born Not applic~
```

```
#add region and development group
simulated_groups <-
  simulated_ppl |>
  left_join(country_region, by = "country") |>
```

```
  mutate(
    development_group = case_match(
      region,
      c("Northern America", "Oceania", "Western Europe", "Northern Europe") ~ "More developed",
      c("Latin America", "Southern Europe", "Eastern Europe") ~ "Less developed",
      .default = NA_character_
    )
  )

head(simulated_groups)
```

```
# A tibble: 6 x 7
  country sex    education      birth_place arrival_age region development_group
  <chr>   <chr>  <chr>          <chr>       <chr>       <chr>  <chr>
1 Canada  Female Below upper s~ Foreign-bo~ Arrived be~ North~ More developed
2 Canada  Female Below upper s~ Foreign-bo~ Arrived at~ North~ More developed
3 Canada  Female Below upper s~ Native-born Not applic~ North~ More developed
4 Canada  Female Upper seconda~ Foreign-bo~ Arrived be~ North~ More developed
5 Canada  Female Upper seconda~ Foreign-bo~ Arrived at~ North~ More developed
6 Canada  Female Upper seconda~ Native-born Not applic~ North~ More developed
```

```
#add unemployment rate
simulated_rates <-
  simulated_groups |>
  mutate(
    unemployment_rate = sample(seq(1, 15, by = 0.1),
                               size = n(),
                               replace = TRUE)
  )

head(simulated_rates)
```

```
# A tibble: 6 x 8
  country sex    education      birth_place arrival_age region development_group
  <chr>   <chr>  <chr>          <chr>       <chr>       <chr>  <chr>
1 Canada  Female Below upper s~ Foreign-bo~ Arrived be~ North~ More developed
2 Canada  Female Below upper s~ Foreign-bo~ Arrived at~ North~ More developed
3 Canada  Female Below upper s~ Native-born Not applic~ North~ More developed
4 Canada  Female Upper seconda~ Foreign-bo~ Arrived be~ North~ More developed
5 Canada  Female Upper seconda~ Foreign-bo~ Arrived at~ North~ More developed
6 Canada  Female Upper seconda~ Native-born Not applic~ North~ More developed
# i 1 more variable: unemployment_rate <dbl>
```

```r
simulated_rates |>
  summarise(min_rate =
            min(unemployment_rate),
            max_rate = max(unemployment_rate))
```

```
# A tibble: 1 x 2
  min_rate max_rate
     <dbl>    <dbl>
1        1     14.9
```

```r
simulated_data <-
  simulated_rates |>
  select(region, development_group, sex, education, birth_place, arrival_age, unemployment_ra
```

```r
women_tertiary_simulated <-
  simulated_data |>
  filter(sex == "Female",
         education == "Tertiary") |>
  drop_na(development_group, unemployment_rate) |>
  mutate(comparison_group = if_else(birth_place == "Native-born",
                                    "Native-born",
                                    arrival_age))

#only foreign-born women for the first mean
mean_by_group <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(development_group) |>
  summarise(mean_rate = mean(unemployment_rate))

mean_by_group
```

```
# A tibble: 2 x 2
  development_group mean_rate
  <chr>                <dbl>
1 Less developed        7.33
2 More developed        8.33
```

```r
#mean rate of native-born women
native_born_mean <-
```

```
  women_tertiary_simulated |>
  filter(birth_place == "Native-born") |>
  group_by(region, development_group) |>
  summarise(native_born_rate = mean(unemployment_rate), .groups = "drop")

native_born_mean
```

```
# A tibble: 7 x 3
  region           development_group native_born_rate
  <chr>            <chr>                        <dbl>
1 Eastern Europe   Less developed                6.05
2 Latin America    Less developed                6.75
3 Northern America More developed               11.6
4 Northern Europe  More developed                7.99
5 Oceania          More developed                4.9
6 Southern Europe  Less developed                7.64
7 Western Europe   More developed                8.43
```

```
#mean rate of foreign-born women, by age at arrival
foreign_born_mean <-
  women_tertiary_simulated |>
  filter(birth_place == "Foreign-born") |>
  group_by(region, development_group, arrival_age) |>
  summarise(foreign_born_rate = mean(unemployment_rate), .groups = "drop")

foreign_born_mean
```

```
# A tibble: 14 x 4
   region           development_group arrival_age              foreign_born_rate
   <chr>            <chr>             <chr>                                <dbl>
 1 Eastern Europe   Less developed    Arrived at age 16 or ol~              9.27
 2 Eastern Europe   Less developed    Arrived before age 16                 6.98
 3 Latin America    Less developed    Arrived at age 16 or ol~              5.58
 4 Latin America    Less developed    Arrived before age 16                 7.72
 5 Northern America More developed    Arrived at age 16 or ol~              7.05
 6 Northern America More developed    Arrived before age 16                 8.05
 7 Northern Europe  More developed    Arrived at age 16 or ol~              8.71
 8 Northern Europe  More developed    Arrived before age 16                 9.21
 9 Oceania          More developed    Arrived at age 16 or ol~             11.2
10 Oceania          More developed    Arrived before age 16                 4.15
11 Southern Europe  Less developed    Arrived at age 16 or ol~              8.28
```

```
12 Southern Europe  Less developed    Arrived before age 16                5.54
13 Western Europe   More developed    Arrived at age 16 or ol~             8.53
14 Western Europe   More developed    Arrived before age 16                7.66
```

```
#calculate the gap by regions
gap_by_region <-
  women_tertiary_simulated |>
  group_by(region, development_group) |>
  summarise(
    native_born_rate = mean(unemployment_rate[birth_place == "Native-born"]),
    foreign_born_before_16_rate = mean(unemployment_rate[arrival_age == "Arrived before age 1
    foreign_born_16_or_older_rate = mean(unemployment_rate[arrival_age == "Arrived at age 16
    .groups = "drop"
  ) |>
  mutate(gap_before_16 = foreign_born_before_16_rate - native_born_rate,
         gap_16_or_older = foreign_born_16_or_older_rate - native_born_rate)

gap_by_region
```

```
# A tibble: 7 x 7
  region           development_group native_born_rate foreign_born_before_16_r~1
  <chr>            <chr>                        <dbl>                       <dbl>
1 Eastern Europe   Less developed                6.05                        6.98
2 Latin America    Less developed                6.75                        7.72
3 Northern America More developed               11.6                         8.05
4 Northern Europe  More developed                7.99                        9.21
5 Oceania          More developed                4.9                         4.15
6 Southern Europe  Less developed                7.64                        5.54
7 Western Europe   More developed                8.43                        7.66
# i abbreviated name: 1: foreign_born_before_16_rate
# i 3 more variables: foreign_born_16_or_older_rate <dbl>, gap_before_16 <dbl>,
#   gap_16_or_older <dbl>
```

```
#calculate the gap by development groups
#The group means average the gaps of regions equally, without weighting by the number of cou
gap_by_development <- gap_by_region |>
  group_by(development_group) |>
  summarise(mean_gap_before_16 = mean(gap_before_16),
            mean_gap_16_or_older = mean(gap_16_or_older))

gap_by_development
```

```
# A tibble: 2 x 3
  development_group mean_gap_before_16 mean_gap_16_or_older
  <chr>                          <dbl>                <dbl>
1 Less developed                -0.0639               0.894
2 More developed                -0.948                0.670
```

Table 1

Table 1 shows 10 randomly selected rows of the simulated data, where each row is a subgroup in one country, defined by region, place of birth and age at arrival, together with its simulated unemployment rate.

```
simulated_data |>
  head(10) |>
  select(region, development_group, birth_place, arrival_age, unemployment_rate) |>
  kable(
    col.names = c("Region", "Group", "Place of birth", "Age at arrival", "Unemployment rate
    caption = "Simulated data"
  )
```

Table 1: Simulated data

Region	Group	Place of birth	Age at arrival	Unemployment rate (%)
Northern America	More developed	Foreign-born	Arrived before age 16	8.8
Northern America	More developed	Foreign-born	Arrived at age 16 or older	9.9
Northern America	More developed	Native-born	Not applicable	11.0
Northern America	More developed	Foreign-born	Arrived before age 16	5.6
Northern America	More developed	Foreign-born	Arrived at age 16 or older	5.2
Northern America	More developed	Native-born	Not applicable	6.7
Northern America	More developed	Foreign-born	Arrived before age 16	9.5
Northern America	More developed	Foreign-born	Arrived at age 16 or older	3.0

Region	Group	Place of birth	Age at arrival	Unemployment rate (%)
Northern America	More developed	Native-born	Not applicable	9.2
Northern America	More developed	Foreign-born	Arrived before age 16	3.1

Table2

Table 2 shows the simulated mean gap in unemployment rate between foreign-born and native-born women in more and less developed regions, separately for women who arrived before age 16 and those who arrived at age 16 or older.

```
gap_by_development |>
  kable(
    digits = 2,
    col.names = c("Group",
                  "Arrived before age 16",
                  "Arrived at age 16 or older"),
    align = c("l", "r", "r"),
    caption = "Simulated mean gap in unemployment rate"
  )
```

Table 2: Simulated mean gap in unemployment rate

Group	Arrived before age 16	Arrived at age 16 or older
Less developed	-0.06	0.89
More developed	-0.95	0.67

PART II: Graph

The following figures use the simulated data to show what the patterns would look like if the data followed each hypothesis.

Figure 1

Figure 1 compare the simulated mean unemployment rate of tertiary-educated foreign-born women in more and less developed regions.

```
#H1a
mean_by_group |>
  ggplot(aes(x = development_group, y = mean_rate)) +
  geom_col() +
  labs(x = NULL,
       y = "Mean unemployment rate",
       title = "Figure 1: Unemployment Rate of Tertiary-Educated\nForeign-Born Women by devel
  theme(plot.title = element_text(size = 12))
```

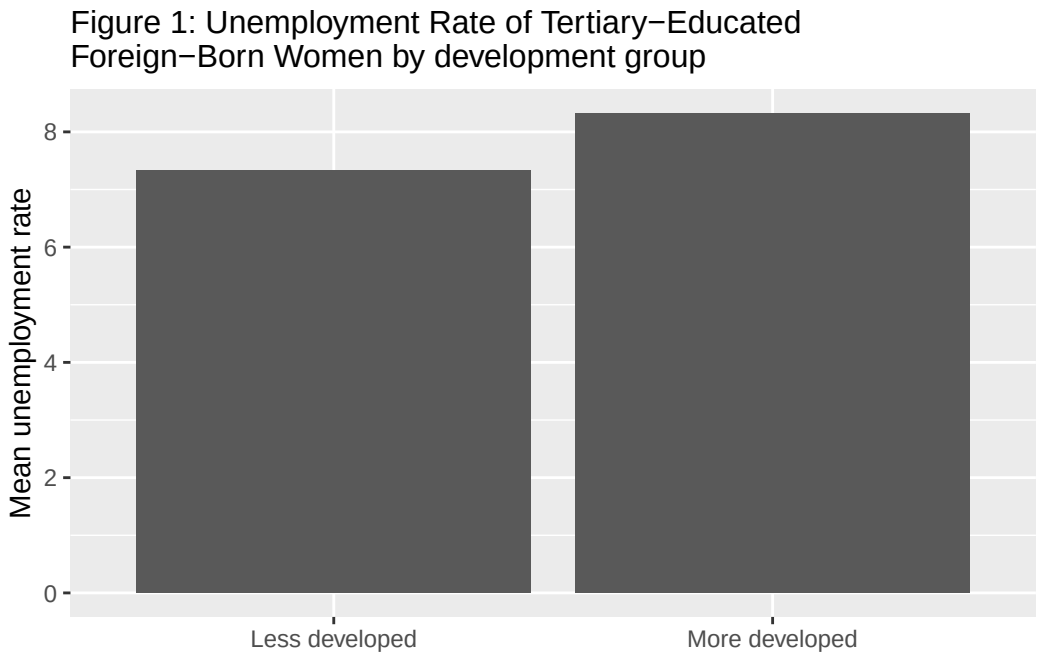


Figure 2

Figure 2 compares the simulated mean unemployment rate of women who arrived before age 16 with those who arrived at age 16 or older.

```
#H2a
mean_by_group_arrival <-
  women_tertiary_simulated |>
```

```
  filter(birth_place == "Foreign-born") |>
  group_by(development_group, arrival_age) |>
  summarise(mean_rate = mean(unemployment_rate),
            .groups = "drop")

mean_by_group_arrival
```

```
# A tibble: 4 x 3
  development_group arrival_age                mean_rate
  <chr>             <chr>                          <dbl>
1 Less developed    Arrived at age 16 or older      7.95
2 Less developed    Arrived before age 16           6.7
3 More developed    Arrived at age 16 or older      8.74
4 More developed    Arrived before age 16           7.92
```

```
mean_by_group_arrival |>
  ggplot(aes(x = development_group, y = mean_rate, fill = arrival_age)) +
  geom_col(position = "dodge") +
  labs(x = NULL,
       y = "Mean unemployment rate",
       fill = "Age at arrival",
       title = "Figure 2: Unemployment Rate of Tertiary-Educated\nForeign-Born Women by Age")
       theme(plot.title = element_text(size = 12))
```

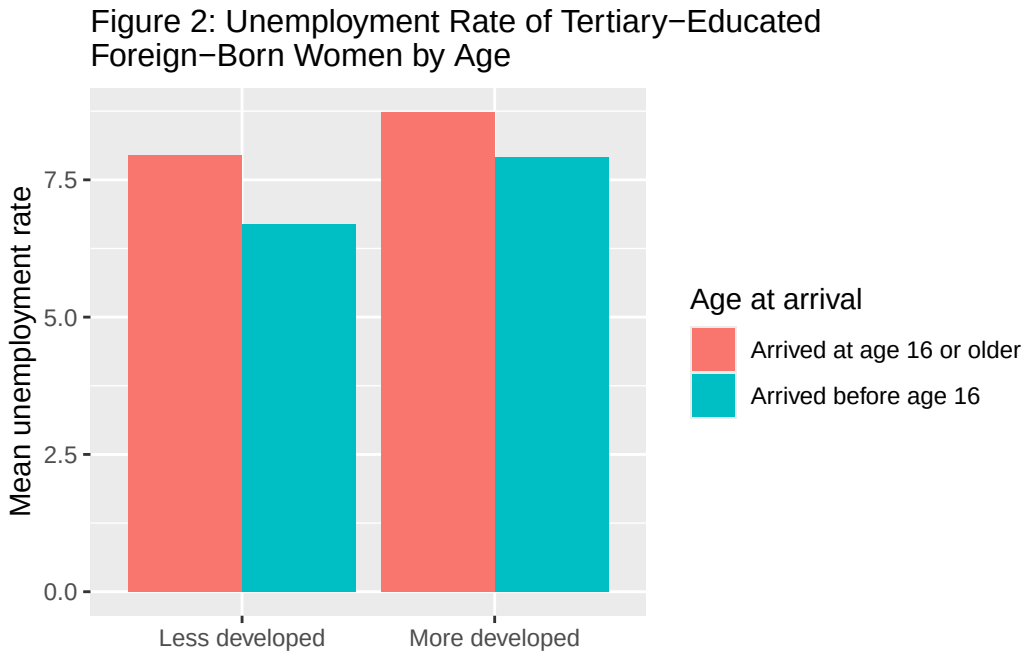


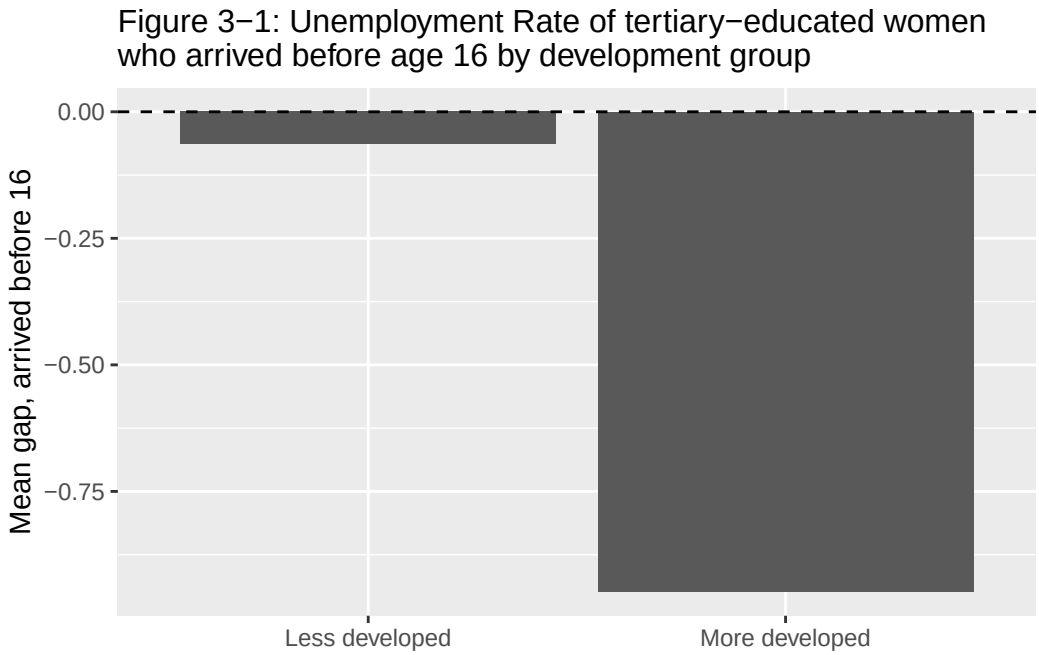
Figure 3

Figure 3-1 compares the simulated mean gap in unemployment rate between foreign-born women who arrived before age 16 and native-born women, in more and less developed regions.

Figure 3-2 shows the same comparison for foreign-born women who arrived at age 16 or older.

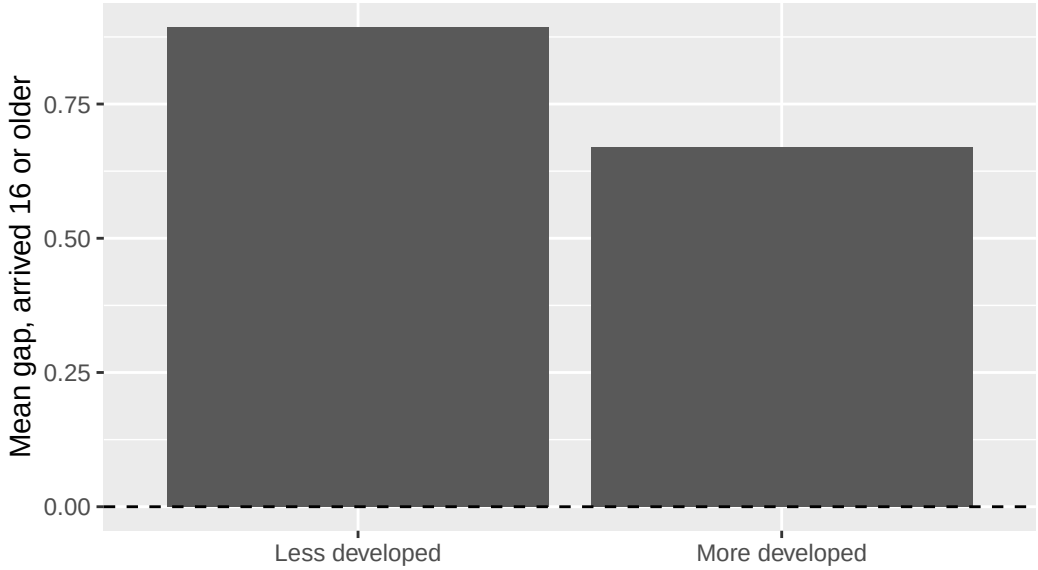
(The group means average the gaps of regions equally, without weighting by the number of countries or by population size.)

```
gap_by_development |>
  ggplot(aes(x = development_group, y = mean_gap_before_16)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = NULL,
       y = "Mean gap, arrived before 16",
       title = "Figure 3-1: Unemployment Rate of tertiary-educated women\nwho arrived before
  theme(plot.title = element_text(size = 12))
```




```
gap_by_development |>
  ggplot(aes(x = development_group, y = mean_gap_16_or_older)) +
  geom_col() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = NULL,
       y = "Mean gap, arrived 16 or older",
       title = "Figure 3-2: Unemployment Rate of tertiary-educated women\nwho arrived at age
  theme(plot.title = element_text(size = 12))
```

Figure 3−2: Unemployment Rate of tertiary−educated women who arrived at age 16 or older by development group



Reference

Alexander, Rohan. 2023. *Telling Stories with Data: With Applications in r and Python*. Chapman & Hall/CRC.

OECD. 2023. *Dataflow DSD_EAG_LSO_EA@DF_LSO_NEAC_UNEMP_MIGR*. OECD Data Explorer; OECD. https://data-explorer.oecd.org.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.